

Abhinandan

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EXPERIENCE

Founding Software Engineer

Sept 2025 – Present

Browser

Remote, San Francisco

- Built Browser's Chrome MV3 recorder and CDP-native browser agent with AX/DOM element capture, cross-iframe support, obstruction checks, and real mouse/keyboard/upload execution.
- Implemented a streaming ReAct runtime across FastAPI + browser extension with SSE tool execution, multi-tab orchestration, safe parallelism, abort/continue semantics, and auditable traces.
- Cut cost per automation run **~3x** across **700K+ monitored production LLM calls** using compact traces, context compression, prompt caching, and dynamic model routing.
- Built a proactive DevRel/content system that generates documentation portals, quickstarts, long-form guides/books, and product demo assets from product workflows, then self-heals affected content as the UI changes.
- Shipped zero-LLM replay for intact workflows with stateful AI fallback, allowing failed automations and generated content flows to resume from the point of failure instead of restarting.

Founding Software Engineer

Jan 2025 – Jul 2025

Cynos Nexus

Noida, India

- Built the AI knowledge base service using FastAPI, LangChain, and vector retrieval pipelines, powering customer-facing search workflows.
- Implemented backend infrastructure for retrieval-augmented AI workflows, including document ingestion, structured data processing, asynchronous workers, Redis-backed queues, and production API serving.
- Productionized AI services with Docker, GitHub Actions, Nginx, cloud deployment, asynchronous processing, and API observability.

Contract Software Engineer

Sept 2024 – Dec 2024

Etkin.ai

Remote, Turkiye

- Built a LangChain + pgvector retrieval system; raised recall@5 from 0.71 → 0.82 on a 300-query labeled eval set.
- Developed scalable data-ingestion pipelines using bulk CSV processing and Celery workers, reducing processing time by 40%

Open Source Contributor

June 2024 – Aug 2024

HeroUI

Y Combinator S24 startup

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after **16 merged open-source contributions**.

SWE Intern

Jan 2024 – Apr 2024

SkilledUp

Noida

- Engineered a context-aware AI code assistant for the core learning dashboard using FastAPI and OpenAI.

OPEN SOURCE PROJECTS

RolloutCore | *PyTorch, NCCL, vLLM* | [GitHub](#)

- Built a RL rollout control plane around vLLM with drain-before-mutate semantics, live NCCL weight installs, post-update cache invalidation, and admit-time version/provenance binding, validated end-to-end without restarting the inference process.
- On Qwen3-8B, measured 4.25s hot updates vs 37.04s full restart ($\sim 8.7\times$ lower model-transition time) on 2x RTX 3090 over PCIe; reproduced 32 stale prefix-cache hits and changed output when cross-version invalidation was deliberately skipped.

MiniServe | *Python, PyTorch* | [GitHub](#)

- Built a from-scratch inference runtime with continuous batching, token-budget scheduling, chunked prefill, recompute preemption, and paged KV with content-addressed prefix caching.
- Implemented a Llama-family path over those paged KV tensors and matched Hugging Face ($\leq 6e-8$ float32 logit error; identical greedy tokens); on a 120-request shared-prefix workload, prefix reuse cut computed prefill work by 97%.

AgentFlow-Pro (Research) | *PyTorch, TRL, DAPO, PRM, LoRA* | [GitHub](#)

- Rebuilt and extended the ICLR 2026 AgentFlow architecture as a local Qwen3-8B Planner → Executor → Verifier → Memory loop with constrained planning, Tavily search, and sandboxed Python/SymPy tools.
- Trained with DAPO + a learned Qwen3-0.6B Process Reward Model and measured **+5.0 pts on GPQA-Diamond** (40.0% → 45.0%) with leakage-free, quantization-matched evaluation.

TECHNICAL SKILLS

ML Engineering: PyTorch, Hugging Face, LangGraph, CrewAI, Qdrant

LLM Systems: Inference engines (vLLM, SGLang, TensorRT-LLM), Fine-tuning (LoRA, QLoRA, PEFT), quantization (AWQ, GPTQ, GGUF)

Engineering: Python, TypeScript, FastAPI, Node.js, Redis, PostgreSQL, AWS, GCP, Docker, GitHub Actions

EDUCATION & RECOGNITION

Dr. APJ Abdul Kalam Technical University

B.Tech, Computer Science | 2026

International Youth Math Challenge Gold Honour | Top 1% TypeScript Engineer Globally (Algora) | Amazon ML

Summer School 2025